



AI for Multimodal Science 2026 @ University of California, San Diego



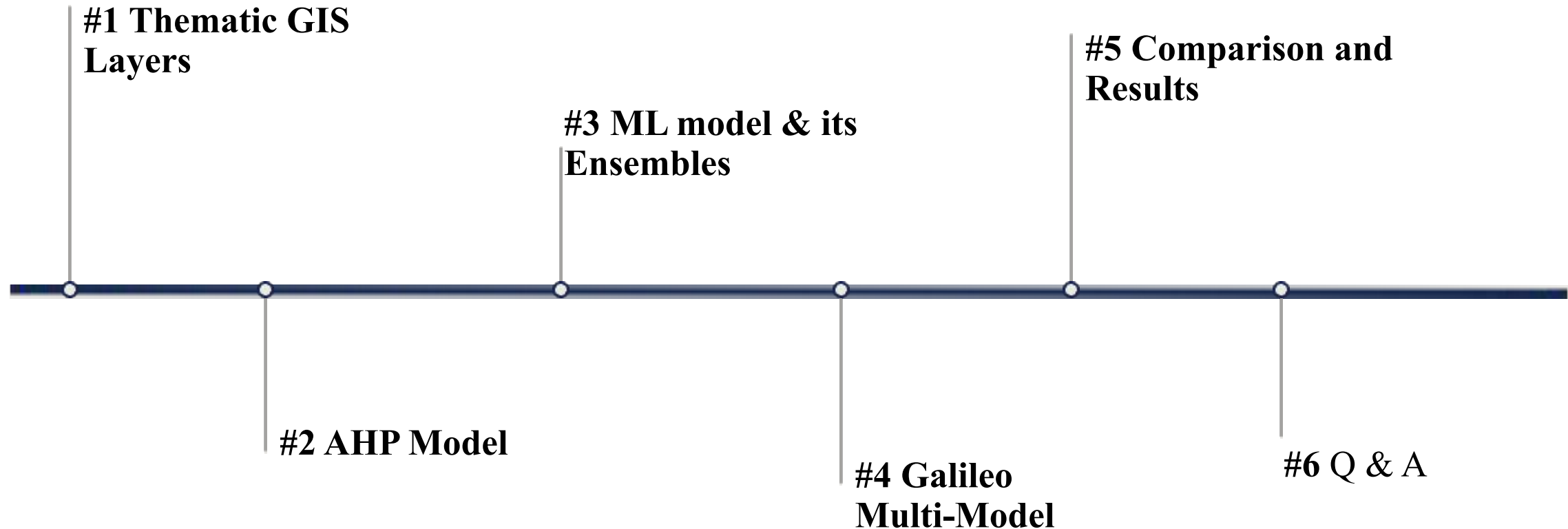
Bridging the Lithological Gap: A Foundation-Model Approach to Groundwater Assessment in Northern Nigeria

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Outline of Presentation



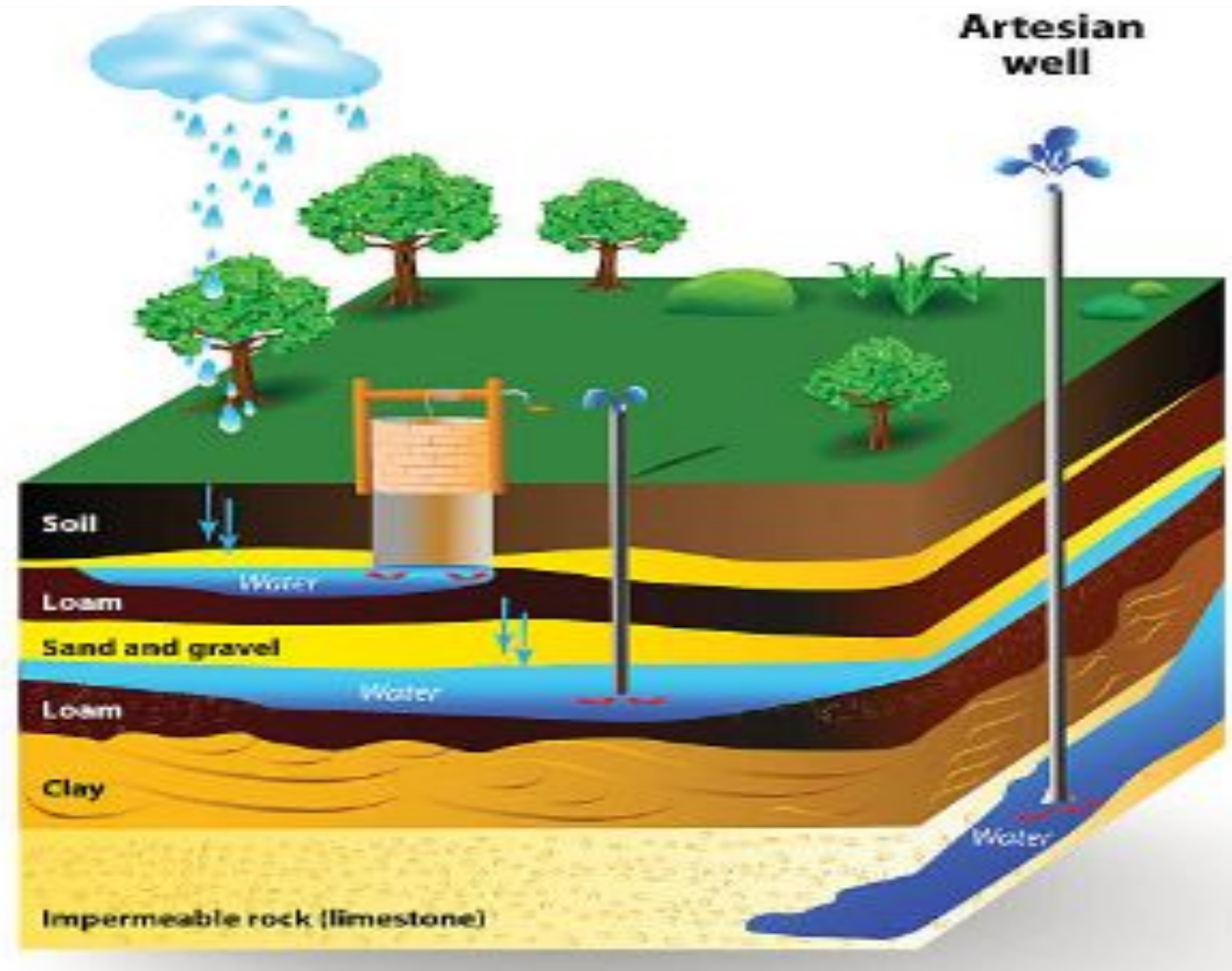
RESEARCH VISION & APPROACH

Challenge: Climate variability and poor resource management threaten water security in Nigeria.

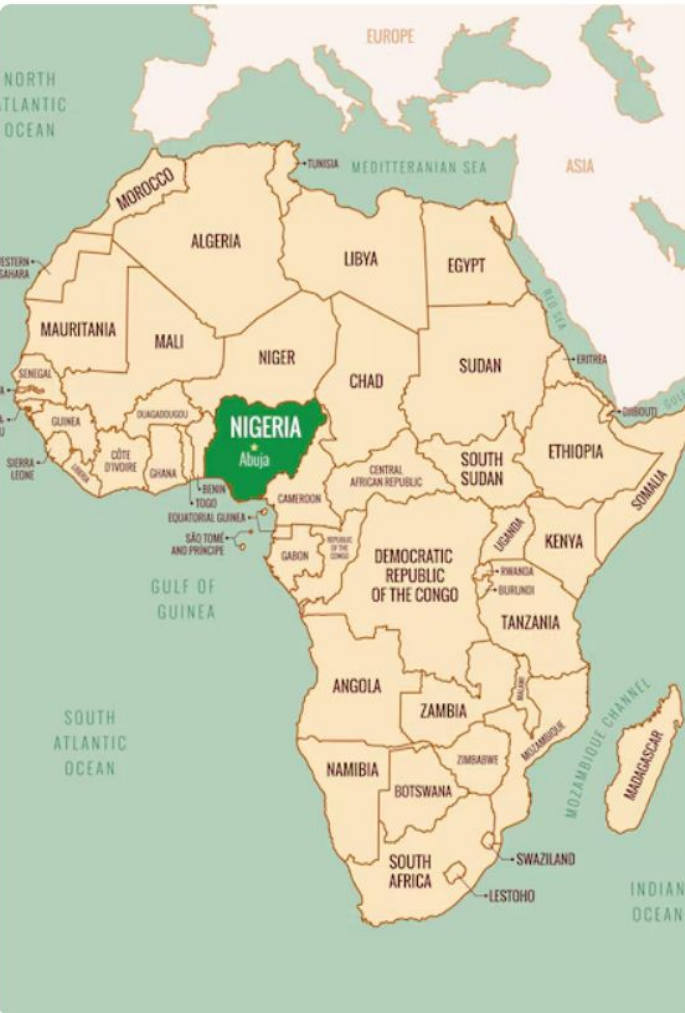
Integration: Combines Geophysics, Remote Sensing, and Machine Learning (ML) for robust analysis.

Goal: Develop an AI-powered predictive model to identify groundwater-favourable zones.

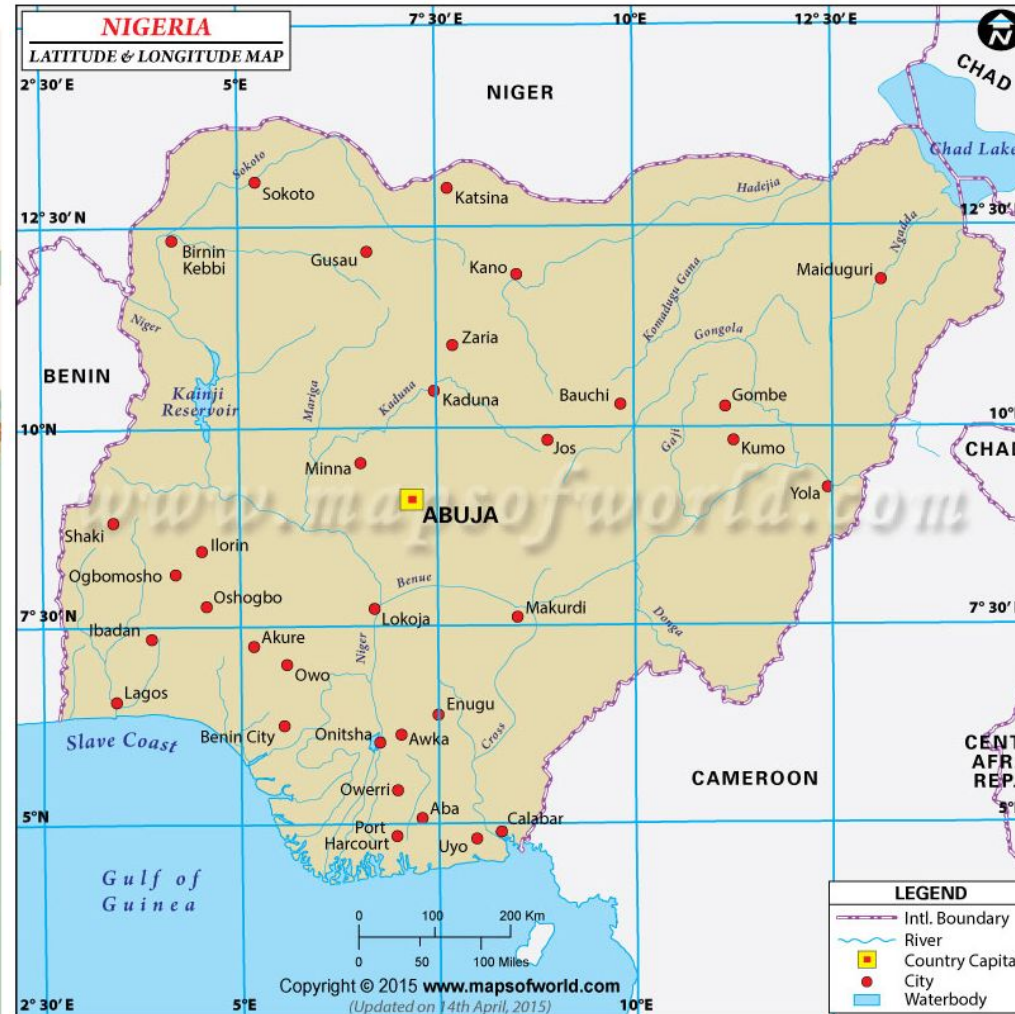
Outcome: A scalable tool guiding groundwater sustainability plan and policy.



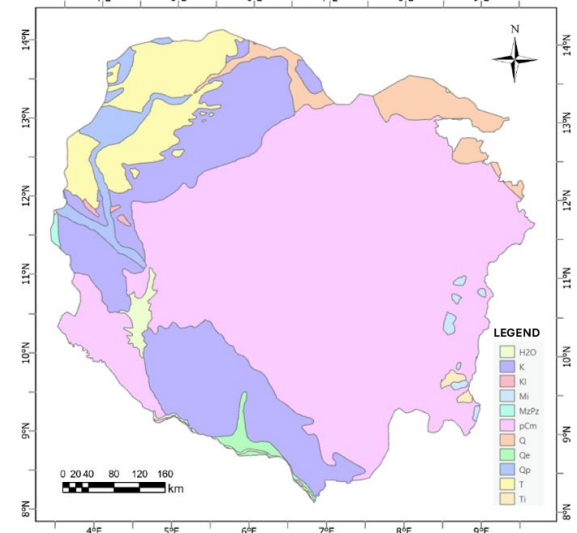
RESEARCH AREA OF INTEREST



African Map

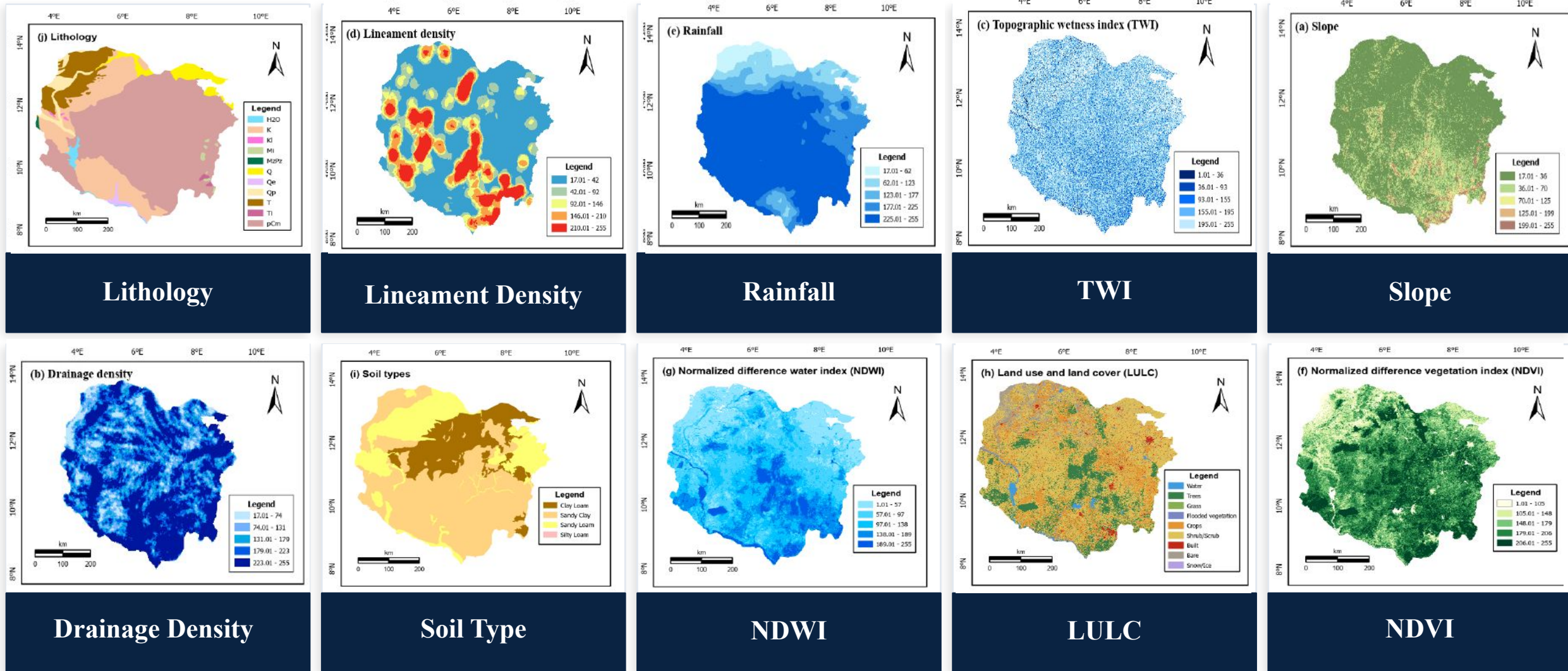


Nigerian Map



Kaduna · Kano · Katsina · Zamfara · Sokoto · Kebbi · Niger/FCT

THEMATIC LAYERS (NW NIGERIA) — 10 GIS INPUT PARAMETERS



Ten thematic remote-sensing and GIS layers used at common resolution

GIS PROCESSING WORKFLOW — ArcGIS Pro

1

Data Acquisition

Sentinel-2 imagery (NDVI, NDWI, LULC)
COPDEM 30m (Slope, TWI, Drainage)
USGS Lithology & Soil datasets
IMERG Rainfall data
Lineament extraction from DEM

2

Coordinate System

Project all rasters to
WGS 1984 UTM Zone 32N
Uniform spatial reference
for all 10 thematic layers

3

Reclassification

Reclassify each layer into
9 groundwater suitability classes
(1=Very Low → 9=Very High)
based on domain knowledge

4

AHP Weighting

Apply Saaty pairwise weights
to each reclassified raster
CR < 0.10 verified
Priority vector normalised

5

Weighted Overlay

Combine all 10 weighted layers
using Raster Calculator:
 $GWPZ = \sum(W_i \times R_i)$
Output: GWP raster map

6

GWP Zones

Classify final index into:
Very High / High / Moderate
Low / Very Low potential
Export map for validation

Priorities

These are the resulting weights for the criteria based on your pairwise comparisons:

Cat		Priority	Rank	(+)	(-)
1	Lithology	29.4%	1	10.9%	10.9%
2	Lineament	21.5%	2	6.8%	6.8%
3	Rainfall	15.4%	3	4.5%	4.5%
4	Drainage	10.9%	4	3.2%	3.2%
5	Slope	7.6%	5	2.3%	2.3%
6	TWI	5.4%	6	1.7%	1.7%
7	Soil	3.8%	7	1.2%	1.2%
8	LULC	2.7%	8	0.9%	0.9%
9	NDVI	1.9%	9	0.7%	0.7%
10	NDWI	1.5%	10	0.7%	0.7%

Number of comparisons = 45
Consistency Ratio CR = 3.2%

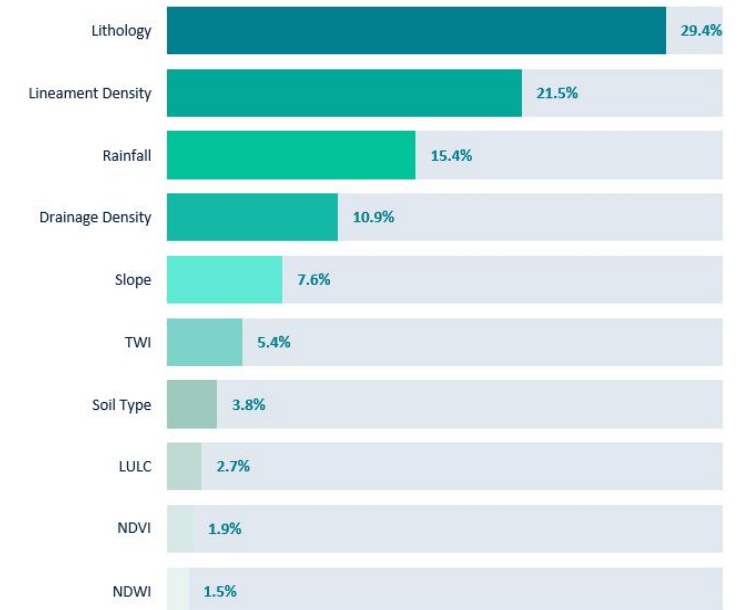
Decision Matrix

The resulting weights are based on the principal eigenvector of the decision matrix:

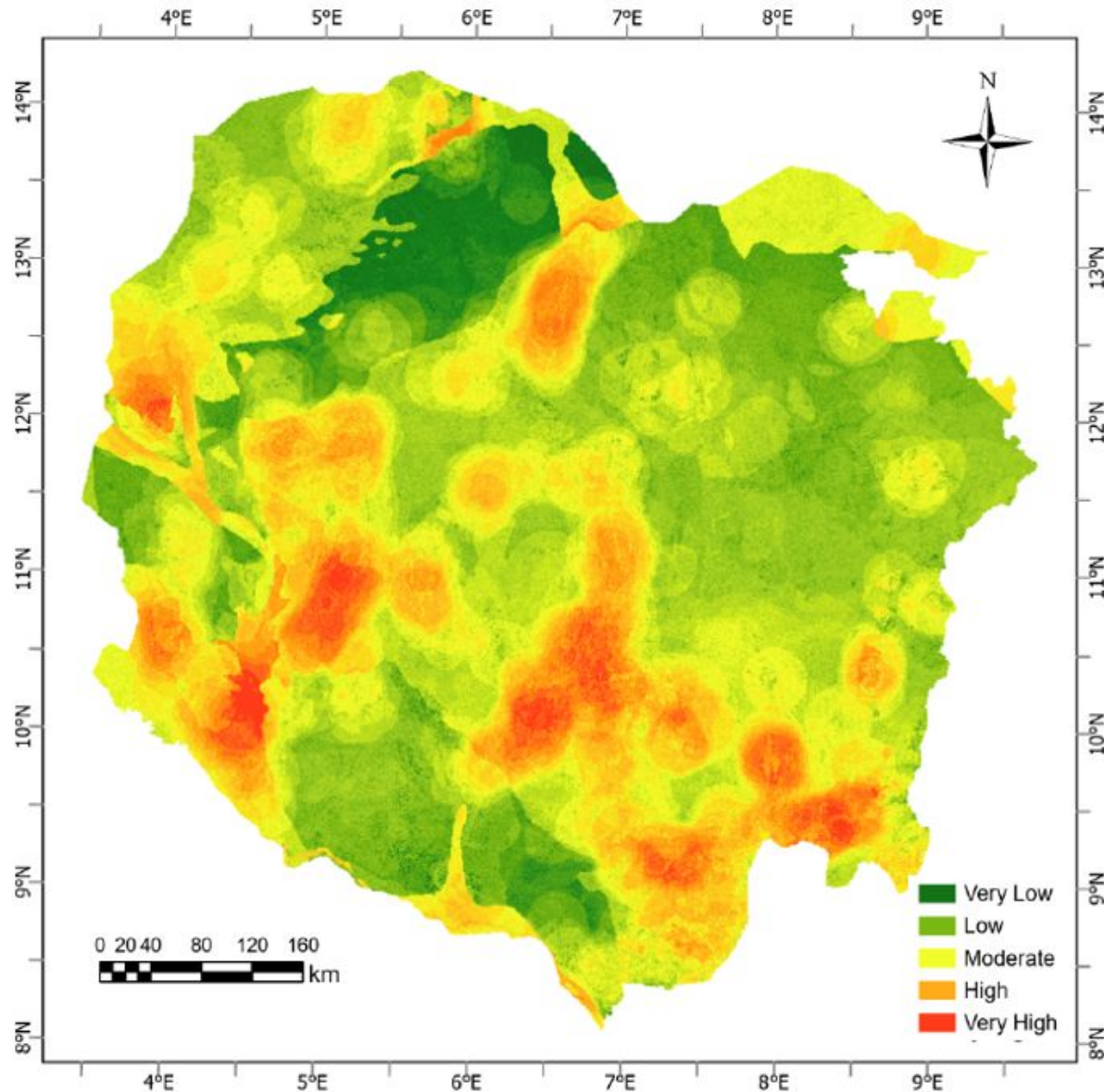
	1	2	3	4	5	6	7	8	9	10
1	1	2.00	3.00	4.00	5.00	6.00	7.00	8.00	9.00	9.00
2	0.50	1	2.00	3.00	4.00	5.00	6.00	7.00	8.00	9.00
3	0.33	0.50	1	2.00	3.00	4.00	5.00	6.00	7.00	8.00
4	0.25	0.33	0.50	1	2.00	3.00	4.00	5.00	6.00	7.00
5	0.20	0.25	0.33	0.50	1	2.00	3.00	4.00	5.00	6.00
6	0.17	0.20	0.25	0.33	0.50	1	2.00	3.00	4.00	5.00
7	0.14	0.17	0.20	0.25	0.33	0.50	1	2.00	3.00	4.00
8	0.12	0.14	0.17	0.20	0.25	0.33	0.50	1	2.00	3.00
9	0.11	0.12	0.14	0.17	0.20	0.25	0.33	0.50	1	2.00
10	0.11	0.11	0.12	0.14	0.17	0.20	0.25	0.33	0.50	1

Principal eigen value = 10.424
Eigenvector solution: 5 iterations, delta = 6.8E-8

AHP Criteria Weights (Saaty Pairwise Comparison)



RESULTS — AHP GROUNDWATER POTENTIAL MAP



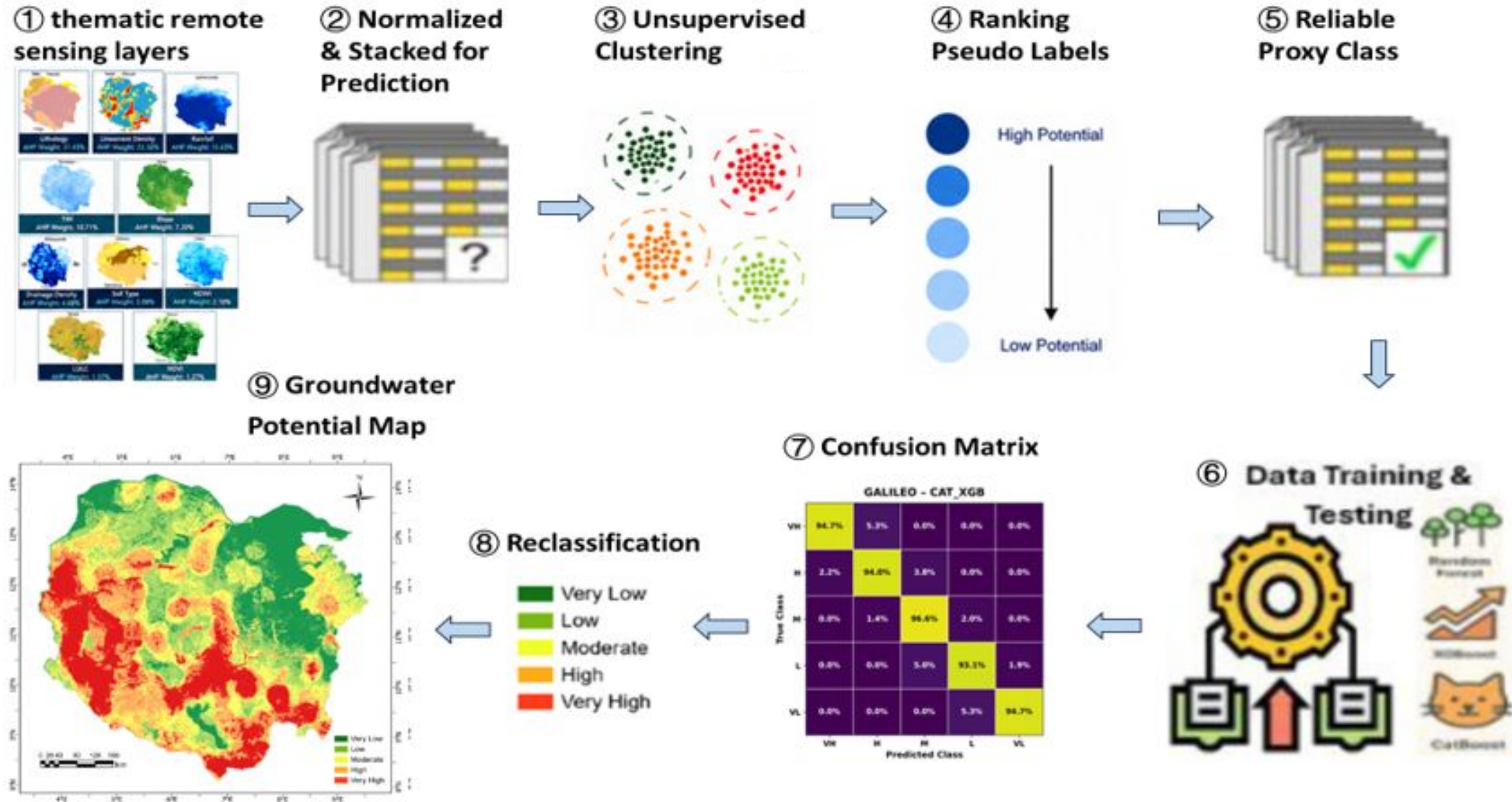
AHP Result Explained

- ✓ CR = 0.032 – Strong reliability
- ✓ Approximately 45-60% AOI is Low to Very Low (Over the Basement Complex)
- ✓ The High to Very High at SW and NW (Sokoto Basin) Favorable for Hydrogeological conditions
- ✓ High zones priority Target for GW development
- ✓ Low zones could be managed for aquifer recharged interventions
- ✓ Result aligns with regional geology with reliable basis for planning obvious hydrogeological boundaries

Facts

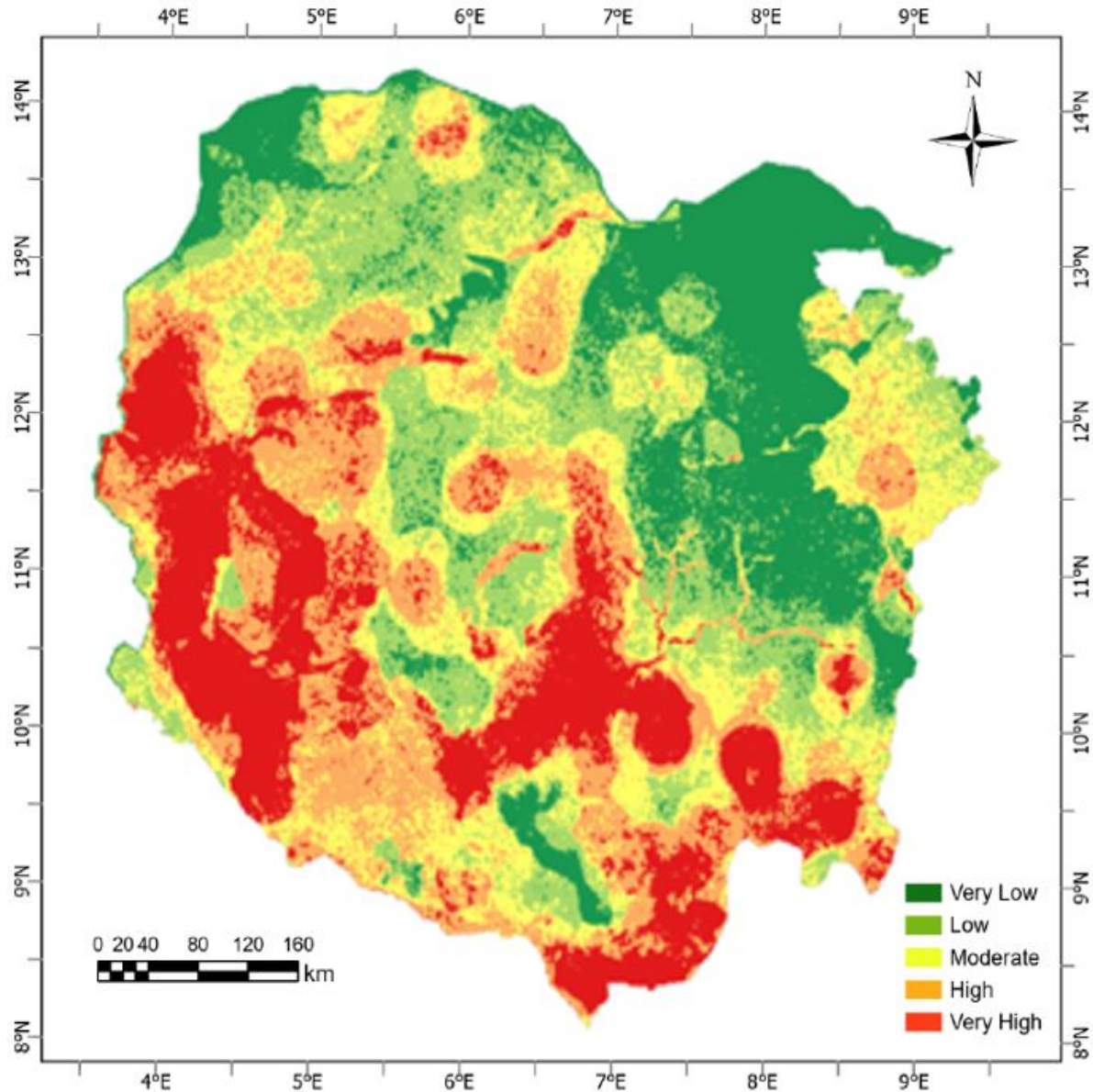
Thematic Layers = 10
CR = 0.032
No Ground Truth Yet

ML MULTIMODAL FRAMEWORK



Error Metric: Acc:0.954, F1:0.955, Prec:0.955, Rec:0.954, Kappa:0.943

RESULTS —ML (CAT_XGB) GROUNDWATER POTENTIAL MAP



ML Advantages

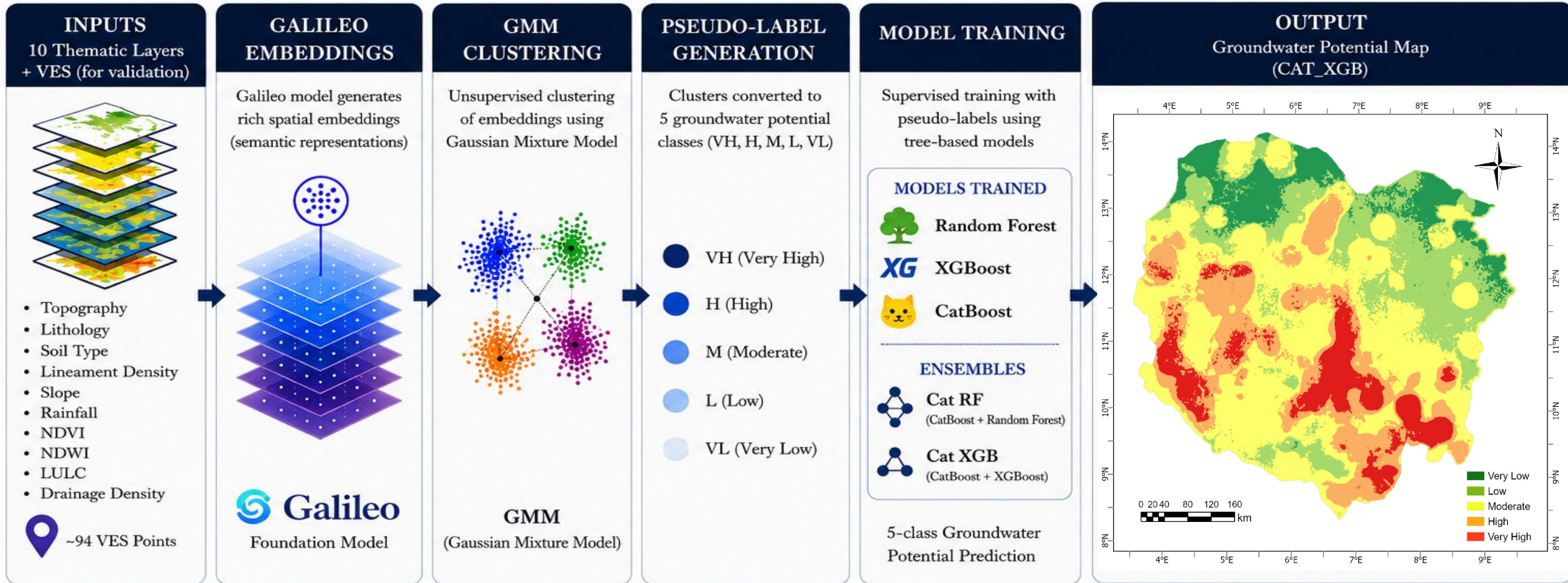
- ✓ It has more intense/extensive red (Very High) areas, especially in the southwest and central zones
- ✓ Shows sharper, more fragmented high-potential clusters
- ✓ Result is more optimistic/aggressive in predicting high potential areas

Recommended Models

CATB-XGB: Better Acc-95.4%, F1-95.5%

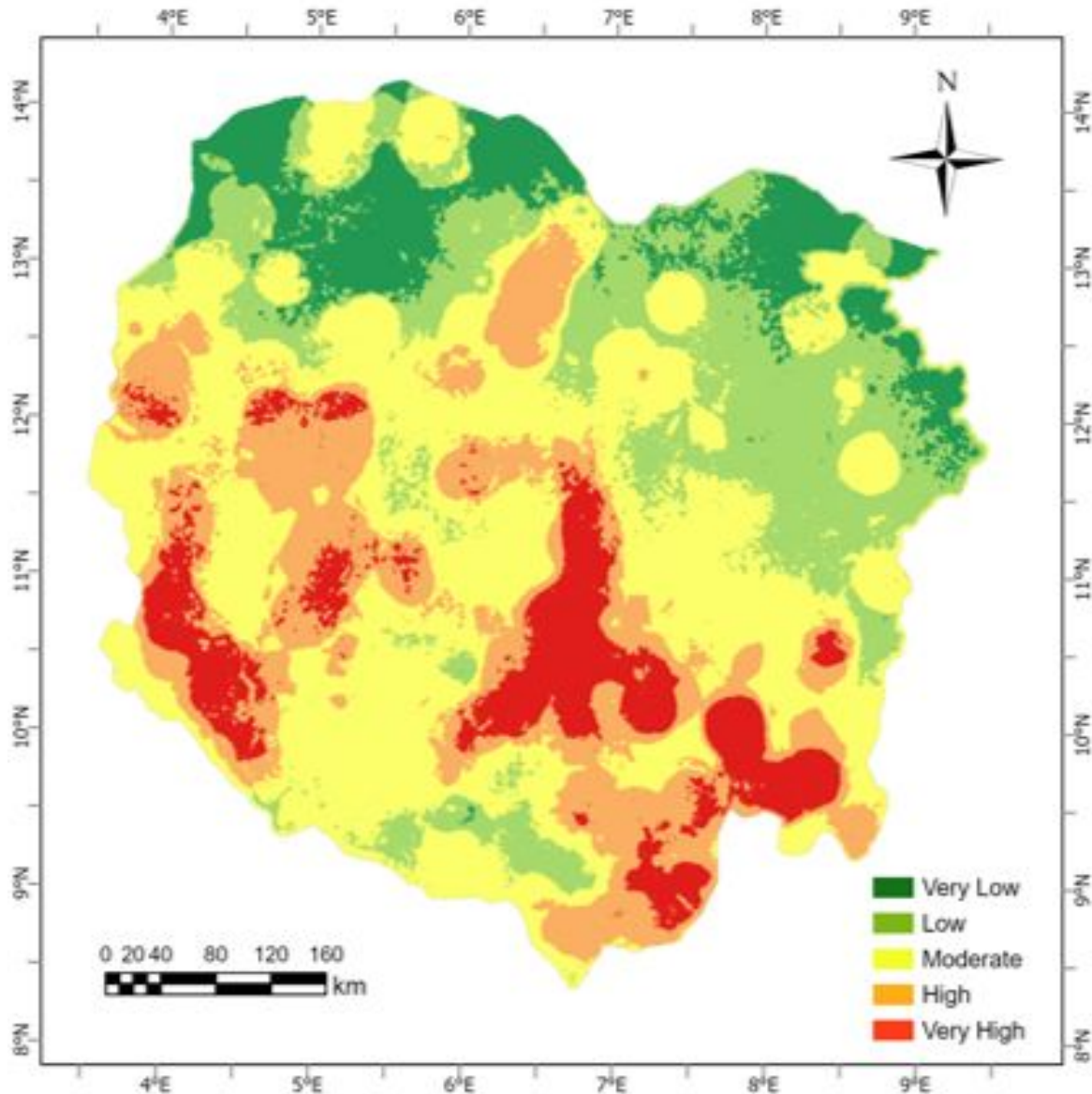
Acc:0.954, F1:0.955, Prec:0.955, Rec:0.954,
Kappa:0.943

GALILEO MULTI-MODAL FRAMEWORK



Error Metrix: Acc:0.950, F1:0.948, Prec:0.950, Rec:0.946, Kappa:0.985

RESULTS —GMM (CAT_XGB) GROUNDWATER POTENTIAL MAP



Galileo Advantages

- ✓ It appears slightly smoother with more balanced yellow/moderate and green coverage, showing less extreme red patches
- ✓ The map has softer transitions and slightly larger contiguous moderate zones
- ✓ It is a more conservative/smooth

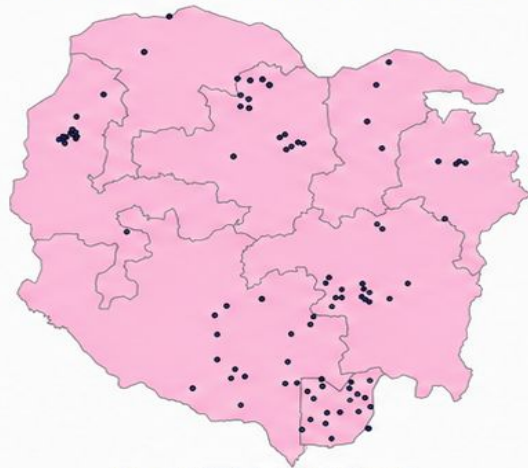
Recommended Models

CATB-XGB: Better Acc-95.4%, F1-95.5%

Acc: 0.950, F1: 0.948, Prec: 0.950, Rec: 0.946,
Kappa: 0.985

VES VALIDATION AND RESULTS

A. VES VALIDATION WORKFLOW



● ~94 VES Points Collected Across Northwestern Nigeria



Extract Model Predictions at VES Locations



Compare with VES Inferred Classes



Evaluate Model Performance

B. CONFUSION MATRIX (GALILEO - CAT_XGB)

		Predicted Class				
		VH	H	M	L	VL
True Class	VH	45	3	0	0	0
	H	2	40	2	0	0
	M	1	3	39	2	1
	L	0	1	4	41	1
	VL	0	0	1	2	45

(Total VES Points = 94)

C. PERFORMANCE METRICS



Accuracy

0.90



F1-Score (Macro)

0.89



Kappa Coefficient

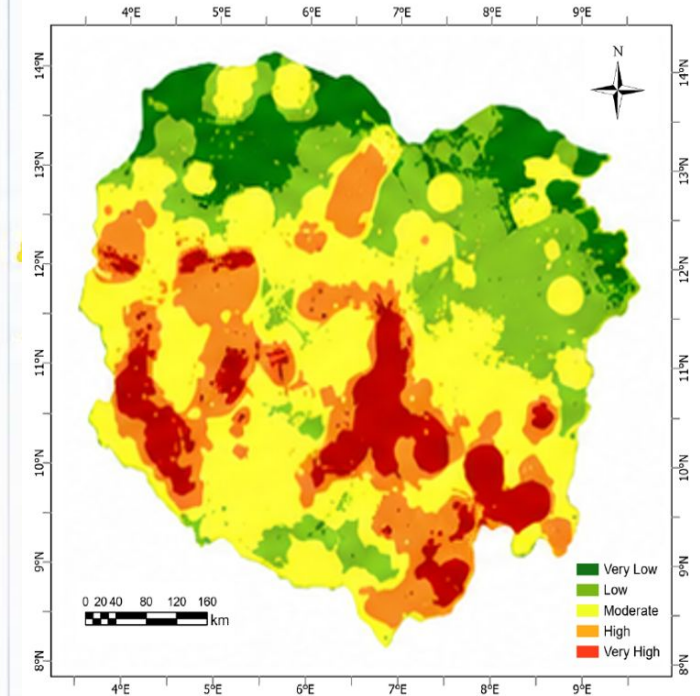
0.87



ROC / AUC (OvR)

0.94

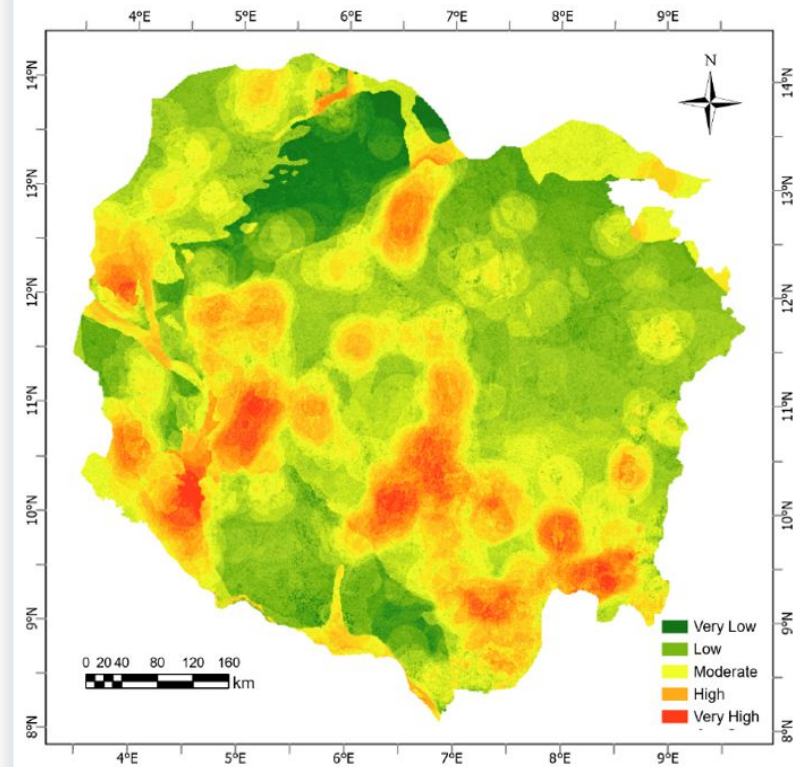
D. RESULT



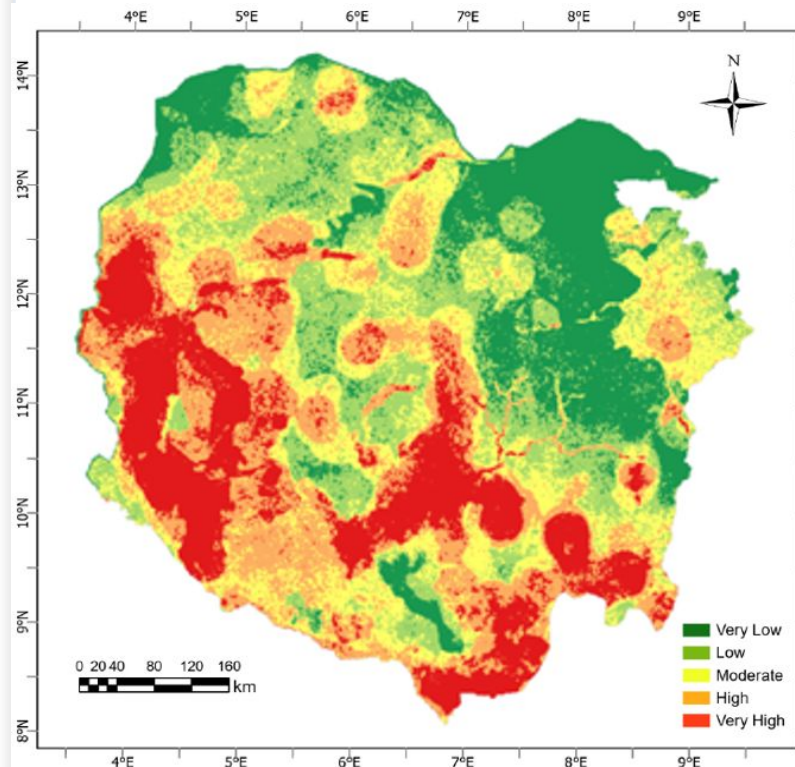
Error Metrix: Acc:0.988, F1:0.987, Prec:0.985, Rec:0.990, Kappa:0.938

COMPARISON OF GWP MAPS

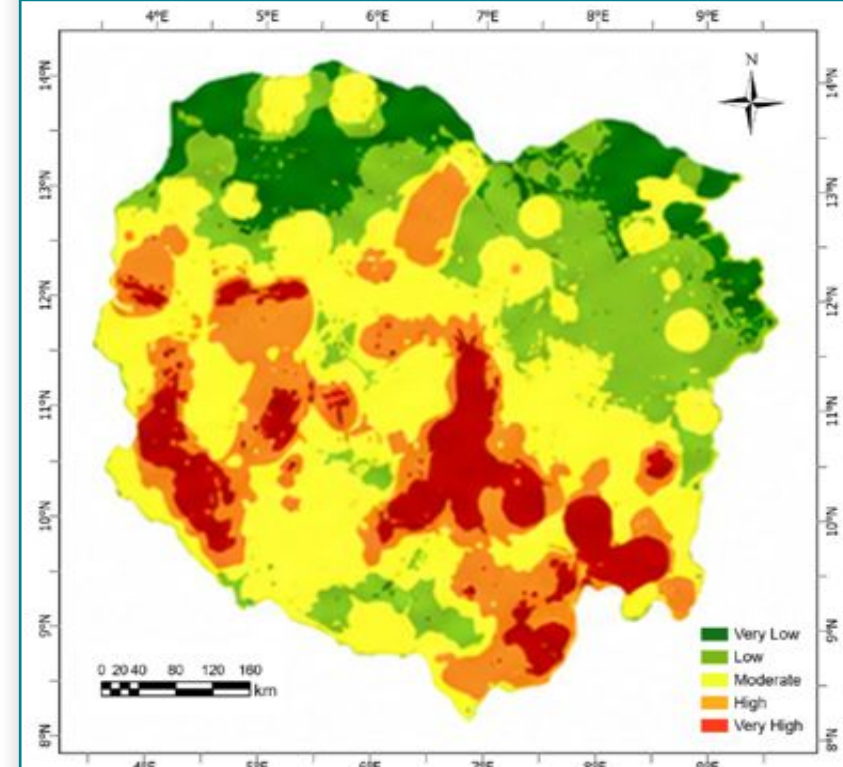
Method 1: AHP



Method 2: Raw ML (CATB)



Method 3: Galileo MM



Criterion

AHP

Raw ML (CATB)

Galileo ML (CAT_XGB)

Data requirement

GIS layers only

GIS layers + VES + ML- Derived labels

GIS layers + VES + GMM information

Spatial coherence

Moderate

Lower (noisy patches)

High (smooth zones, geologically coherent)

Ground truth needed

Expert judgement only (No VES)

VES used for model checking

VES used as geophysical guide & for validation

Role in this study

Policy oriented baseline map

ML Baseline reference

Preferred product for detail GW mapping

CONCLUSIONS

Multi-method validation

01

Integrating AHP, raw ML, and Galileo-ML produces a robust groundwater potential framework with cross-validation through VES borehole data.

AHP remains valuable

02

Saaty pairwise weighting with 10 GIS layers effectively delineates potential zones with expert-guided transparency and interpretability.

Galileo ML performance

03

The Galileo spatial embedding approach (CAT_XGB) produces smoother, geologically more realistic groundwater potential maps compared to pixel-based ML.

Key drivers identified

04

Lithology and lineament density dominate groundwater potential, guiding future borehole siting priorities in Northwestern Nigeria.

THE RESEARCH GROUP



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Combining geophysics, remote sensing and spatial AI enables more reliable groundwater exploration in data-scarce regions.

Q&A

Thank you!

